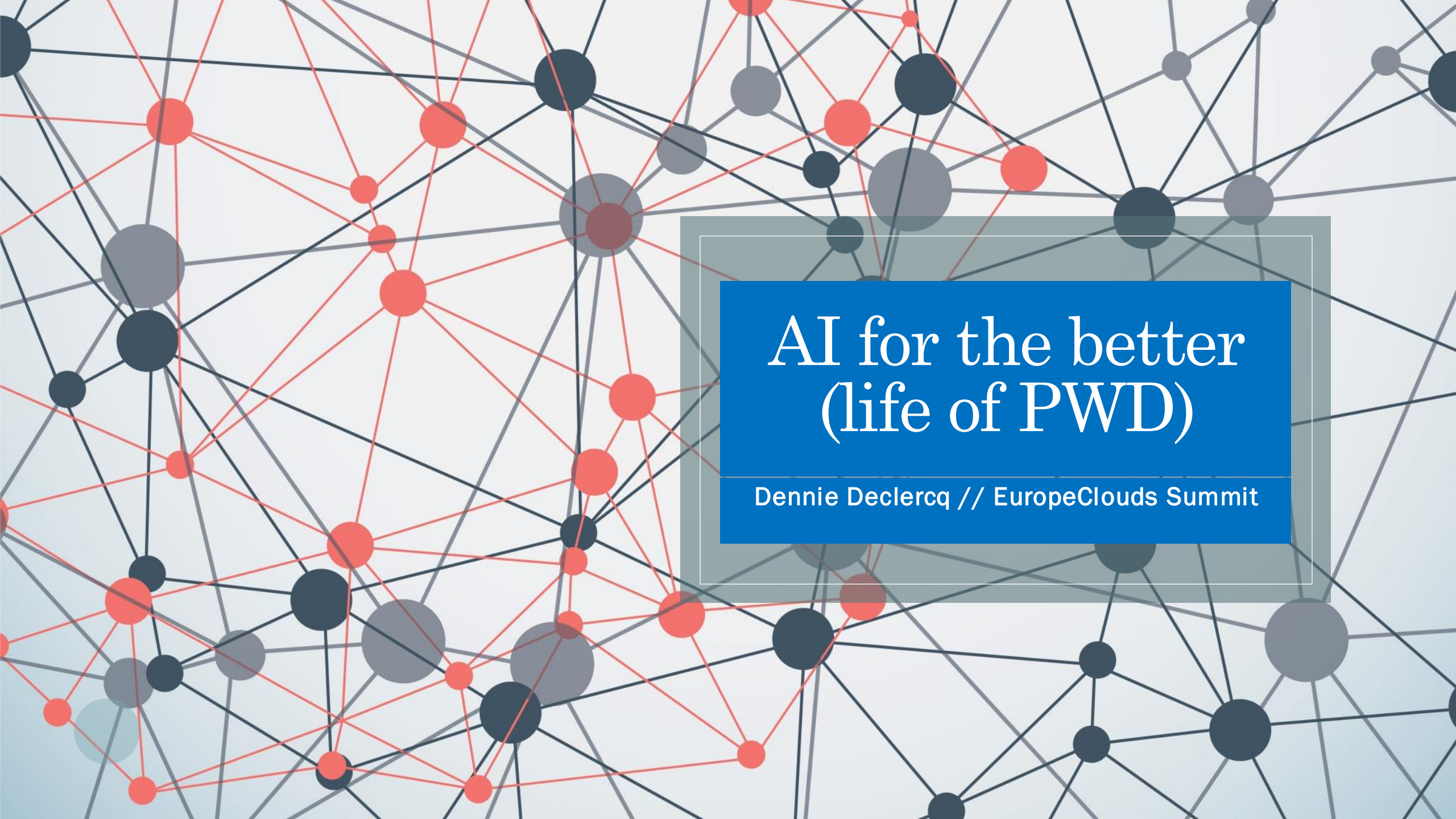


A complex network diagram with nodes of various sizes and colors (black, red, grey) connected by lines, creating a web-like structure. A blue rectangular box is overlaid on the right side of the image, containing the title and speaker information.

AI for the better (life of PWD)

Dennie Declercq // EuropeClouds Summit



Hi, My name is!

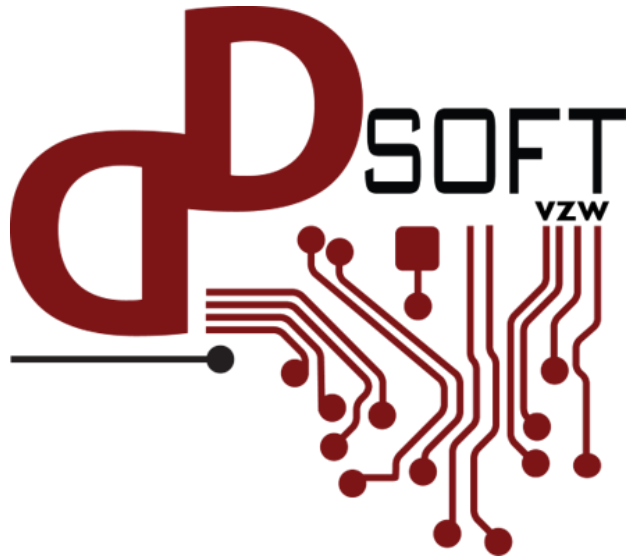
- Dennie Declercq
- @DennieDeclercq
- #DreamingIsBelieving
- President & Developer DDSoft

- NOT an AI engineer 😊





 Microsoft
Cognitive Services



And I am NOT an AI
engineer

Content

- Primer into Accessibility
- Microsoft Cognitive Services API overview
- Real world problems
- The real inspiration
- Real world inspiration
- AI For Good
- How to get started?
- Wrap up
- Thanks || Q&A



IMAGINE



Primer into Accessibility

Kind of disabilities



Learning
disability



Autism



Visual



Auditory



Motoric



Much more...

Learning disability?

Significant limitations in both:

- Intellectual functioning
- Adaptive behavior

Learning Disability

Intellectual Functioning

Also called intelligence

Mental capacity

IQ score between 70-75

Adaptive Behavior

Conceptual Skills

- Language/literacy
- Money
- Time/numbers

Social Skills

- Social responsibility
- Self-esteem

Practical skills

- Activities daily living
- Personal care / healthcare

Principles of Inclusion



- Same activities
- Same district
- Work Together
- Adjustments par person

Digital Difficulties for PWD.

Learning Disability

- Read
- Understand

Autism

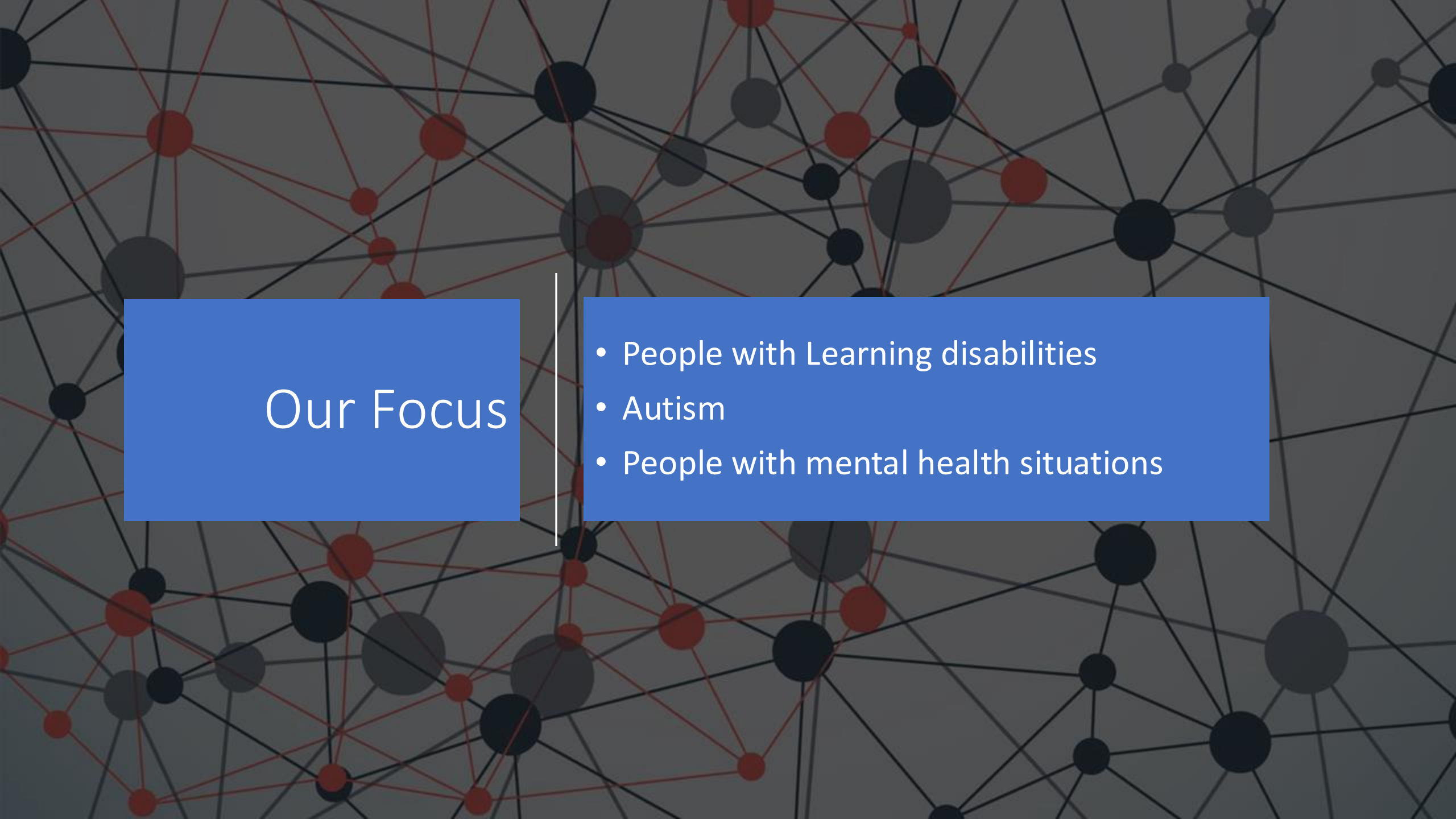
- Understand
- See structure

Blindness or vision loss

- Decreased ability to read
- Decreased ability to navigate

Auditory disabilities

- Hearing notifications
- Following movies without audio description.

The background of the slide is a complex network diagram. It features numerous circular nodes of varying sizes, colored in dark blue, red, and grey. These nodes are interconnected by a dense web of thin, dark grey lines, creating a sense of global connectivity and complexity.

Our Focus

- People with Learning disabilities
- Autism
- People with mental health situations

MICROSOFT COGNITIVE SERVICES API'S

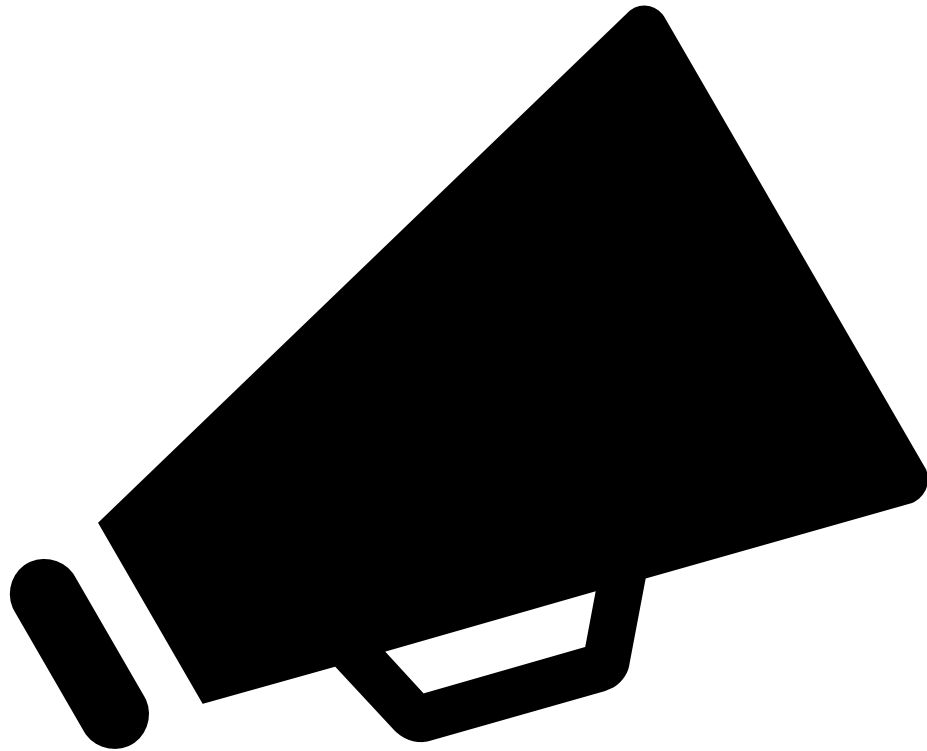
GETTING STARTED WITH COGNITIVE SERVICES

AI + Machine Learning > Cognitive Services

 Bing Autosuggest Microsoft Help users complete queries faster by adding intelligent type-ahead capabilities	 Bing Custom Search Microsoft Create a custom search engine with ad-free results	 Bing Entity Search Microsoft Recognize and classify named entities, and find search results based on them	 Bing Search Microsoft Enable safe, ad-free search for your users with Bing, a leading search engine	 Bing Spell Check Microsoft Help users identify and fix word breaks, slang, names, homonyms, and brands	 Computer Vision Microsoft An AI service that analyzes content in images	 Content Moderator Microsoft An AI service that detects unwanted content
 Custom Vision Microsoft An AI service and end-to-end platform for applying computer vision to your specific scenario	 Face Microsoft An AI service that analyzes faces in images	 Speaker Recognition Microsoft An AI service that enables you to identify individual speakers or use speech as a means of verification	 Speech Microsoft An AI service that enables you to embed human-like speech capabilities into your apps	 QnA Maker Microsoft An AI service that enables you to create a bot from your data in minutes	 Text Analytics Microsoft An AI service that enables you to unlock insights from natural language text using sentiment	 Translator Text Microsoft An AI service that enables you to easily translate text between more than 60 languages
 Personalizer Microsoft An AI service that delivers a personalized user experience	 Language Understanding Microsoft Teach your apps to understand commands from your users	 Ink Recognizer (Preview) Microsoft An AI service that recognizes digital ink content, such as handwriting, shapes, and document layout	 Immersive Reader (preview) Microsoft An AI service that helps users read and comprehend text	 Form Recognizer (preview) Microsoft The AI service that extracts information from documents	 Anomaly Detector (preview) Microsoft An AI service that helps you foresee problems before they occur	 Cognitive Services Microsoft Connect powerful AI to your apps



DEMO: Creating a Cognitive Service on Azure

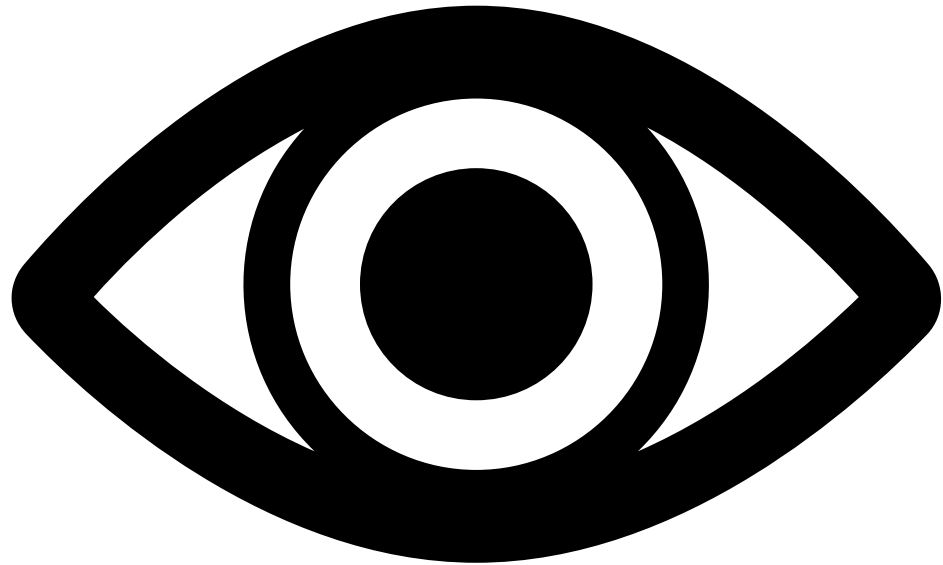


Speech Service

- Text-to-speech
- Custom Voice

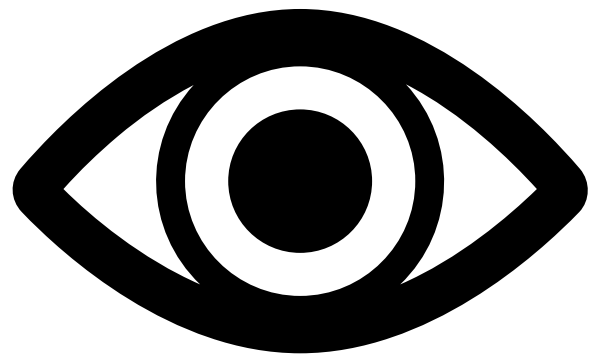


DEMO: Speech API



Vision API

- Analyzes images
- Let PWD take pictures and tells what it is.



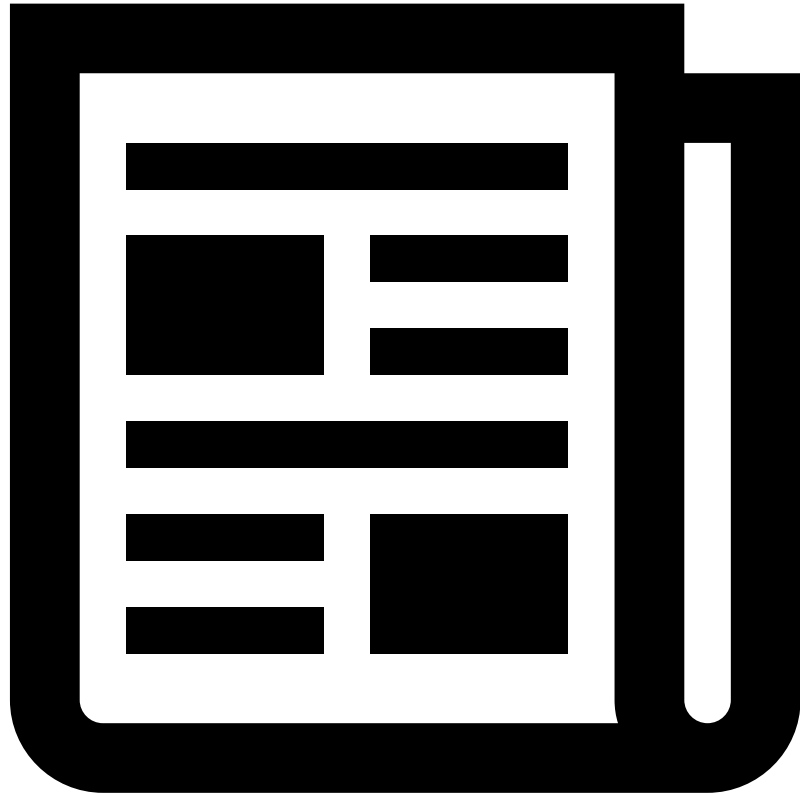
Face API

- Detect Emotion
- Detect Facial Hair
- Detect Smile
- Detect Age



BOT Service

- Provide BOTs
- BOTs really can assist PWD
- Make clear guidelines



Immersive Reader

- Accessibility features like:
 - Reading aloud websites
 - Linking pictographs to websites
 - Text appearance options

PROBLEMS

A thin, vertical white line is positioned to the right of the word "PROBLEMS", extending from the top of the text to the bottom of the text.

Literacy

- Reading information
- Understanding written text
- Recognizing money
- Math



Executive Functions

- Planning
‘simple’ tasks
- Making choices
- Concentration



‘FEELING’

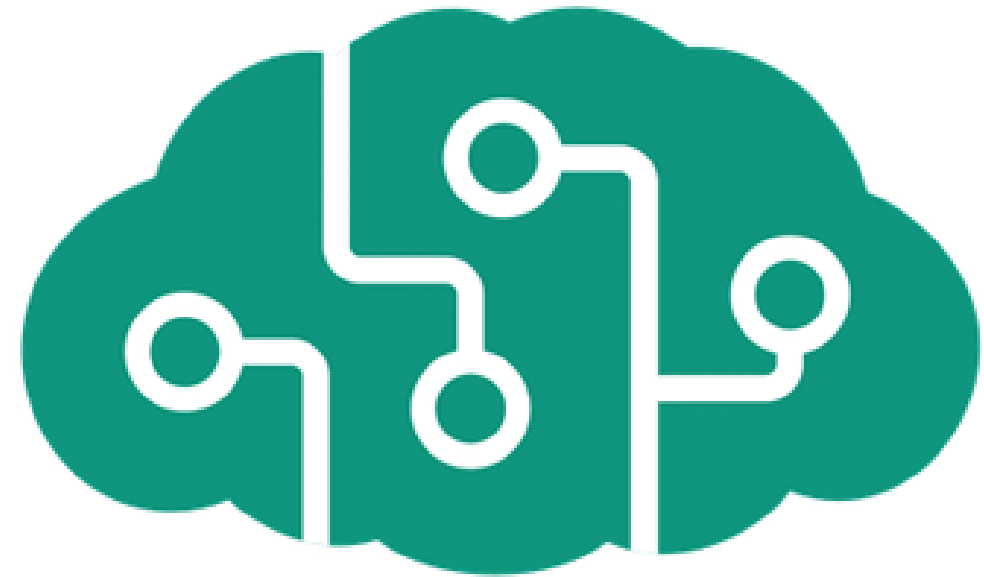
“I have no elbows, I don’t feel if I am going over your borders, and if this annoys you” -Dennie Declercq

Self-trust

- Self-trust can be hard
- Can I really do this?
- Will this really work?



AS YOU
SEE...



Microsoft

Cognitive Services

THE REAL INSPIRATION





^{n.} Learning ^{n.} Tools ^{adj.} Immersive ^{n.} Reader
^{v.} creates a ^{n.} reading experience that
^{v.} adds ^{n.} accessibility and

Learning ^{n.} Tools ^{adj.} Immersive ^{n.} Reader
creates a ^{n.} reading experience that
adds acce  and



The Immersive Recipe



My vision barber

Custom Vision

Face API

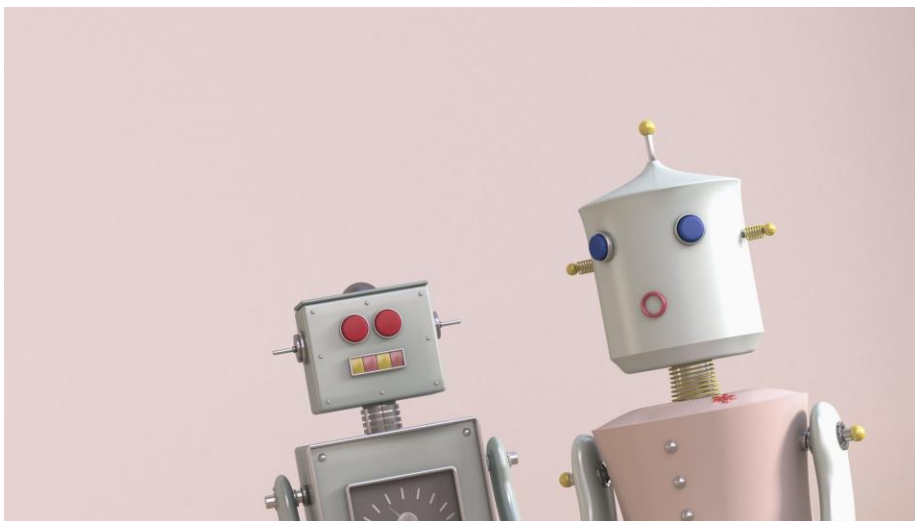
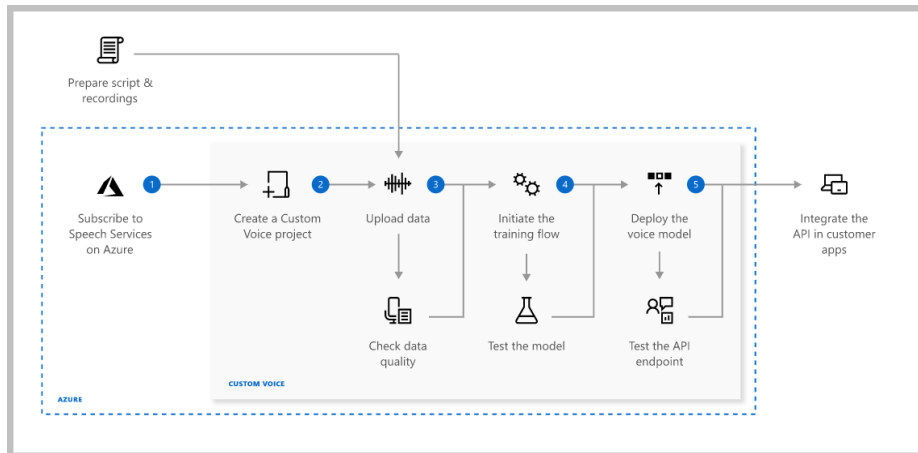
```
[Newtonsoft.Json.JsonProperty(PropertyName="beard")]  
public double Beard { get; set; }
```

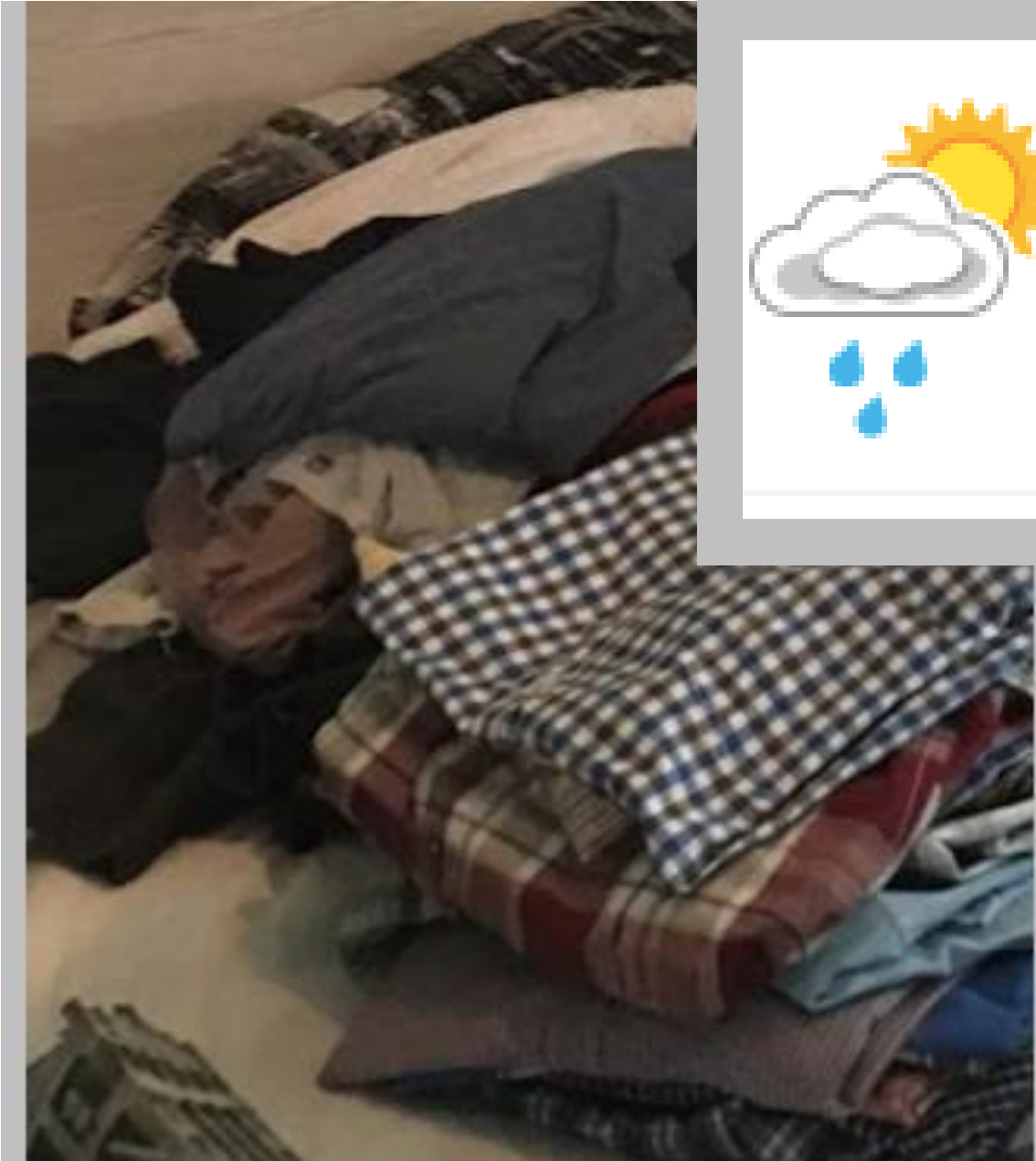
My Artificial Mom

Text To Speech

BOT Service

Custom voice





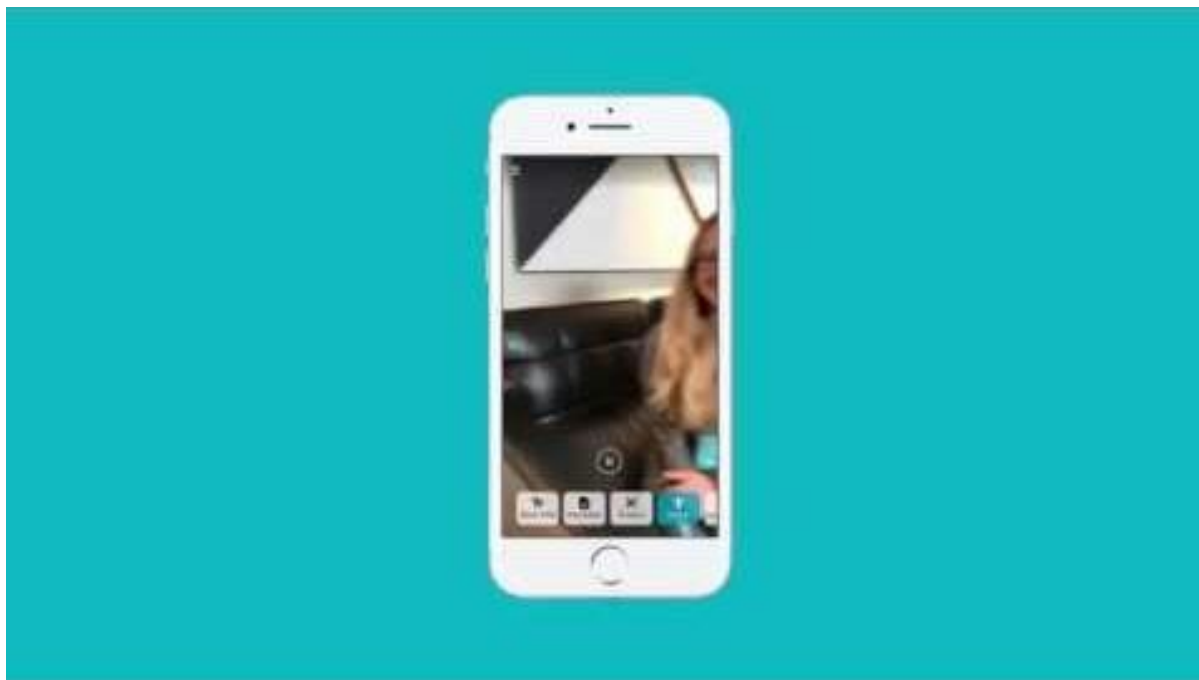
The AI wardrobe

Graph API

Computer Vision

REAL WORLD INSPIRATION

A thin, vertical white line is positioned to the right of the text, extending from the top of the word 'INSPIRATION' to the top of the word 'REAL'.



MICROSOFT SEEING AI
<https://go.ddsoft.be/msftseeingai>



MICROSOFT TRANSLATOR
<https://go.ddsoft.be/msfttranslator>



DEMO Microsoft Translator features bing.com



HELPICTO
<https://go.ddsoft.be/helpicto>

People can add
images or adjust
content as needed



THANK YOU POWERPOINT!
[HTTPS://GO.DDSOFT.BE/AIPPT](https://go.ddssoft.be/aippt)

A man with short brown hair and a beard, wearing a black t-shirt, is sitting at a long wooden table in a modern office. He is looking down at a laptop. On the table, there is a black backpack, a clear plastic cup with a straw, and a black can. The background shows large windows and a wall with a pattern of colorful dots.

AI for Accessibility grants

Empower your work through grants, investments of technology, and expertise.

Apply for a grant >

<https://go.ddsoft.be/msftai>

CALL: AI FOR ACCESSIBILITY

Getting started

1

Do research in
accessibility /
People with
disabilities

2

Learn
Microsoft
Cognitive
Services

3

Get further
inspired

4

Play with
existing tools

WRAP UP

Links & References

- Microsoft Seeing AI: <https://go.ddsoft.be/msftseeingai>
- Helpicto: <https://go.ddsoft.be/helpicto>
- Microsoft Translator movie:
<https://go.ddsoft.be/msfttranslator>
- AI in PowerPoint: <https://go.ddsoft.be/aippt>
- Microsoft AI for Good: <https://go.ddsoft.be/msftai>



Thanks! | | Q&A

- Dennie Declercq
- dennie@ddsoft.be
- @DennieDeclercq
- #DreamingIsBelieving